

INTRODUCCION A LA CRIPTOGRAFIA MODERNA

Homework 3

Depto. de Computación, FCEN, UBA, 1er. Bimestre 2026

28/4/26

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Due: 12/5/26, 23:59 hs

Instructions

- Upload your solution to Campus; make sure it's only one file, and clearly write your name on the first page. Name the file '<your last name>_HW3.pdf.'

If you are proficient with $\text{\LaTeX}$, you may also typeset your submission and submit in PDF format. To do so, uncomment the `"%\begin{solution}"` and `"%\end{solution}"` lines and write your solution between those two command lines.

- Your solutions will be graded on *correctness* and *clarity*. You should only submit work that you believe to be correct.
- You may collaborate with others on this problem set. However, you must **write up your own solutions** and **list your collaborators and any external sources (including ChatGPT and similar generative AI chatbots)** for each problem. Be ready to explain your solutions orally to a member of the course's staff if asked.

This assignment contains 6 questions, for a total of 60 points.

1. (10 points) Consider the following key-exchange protocol:

- (a) Alice chooses uniform $k, r \in \{0, 1\}^n$, and sends $s := k \oplus r$ to Bob.
- (b) Bob chooses uniform $t \in \{0, 1\}^n$, and sends $u := s \oplus t$ to Alice.
- (c) Alice computes $w := u \oplus r$ and sends w to Bob.
- (d) Alice outputs k and Bob outputs $w \oplus t$.

Show that Alice and Bob output the same key. Analyze the security of the scheme (i.e., either prove its security or show a concrete attack).

Solution: Veamos que computan la misma clave.

$$k_A := k$$

$$k_B := w \oplus t = u \oplus r \oplus t = s \oplus t \oplus r \oplus t = s \oplus r = k \oplus r \oplus r = k$$

Veamos ahora que no es seguro. Si un atacante está viendo los mensajes que se intercambian Alice y Bob, puede hacer lo siguiente. En la primera comunicación,

donde Alice le envía s , el adversario se lo guarda. Luego, Cuando Bob envía u , el adversario puede computar $t := s \oplus u$. Finalmente, cuando Bob envía w , el adversario computa $k_B := w \oplus t$. Y así consigue armar la clave de Bob y, por lo tanto, de Alice.

2. (10 points) Formally define the CDH assumption (i.e., first define a CDH experiment, and then use it to define what it means for the CDH problem to be hard). Prove that hardness of the CDH problem relative to $\mathcal{G}$ implies hardness of the discrete-logarithm problem relative to $\mathcal{G}$, and that hardness of the DDH problem relative to $\mathcal{G}$ implies hardness of the CDH problem relative to $\mathcal{G}$.

Solution:

Definimos el siguiente experimento CDH

1. Usar $\mathcal{G}(1^n)$ para obtener $(\mathcal{G}, q, g)$.
2. Elegir x_1, x_2 de manera uniforme.
3. Enviarle al adversario $\mathcal{A}(\mathcal{G}, q, g, g^{x_1}, g^{x_2})$
4. Si el adversario devuelve $g^{x_1 \cdot x_2}$, el experimento termina con resultado 1. Si no, 0.

Definimos CDH de la siguiente manera. El problema CDH es difícil relativo a $\mathcal{G}$ si el experimento descrito arriba tiene probabilidades negligibles de tener éxito. Es decir,

$$Pr(CDH(\mathcal{G}, 1^n)) \leq \text{negl}(n)$$

Ahora, demostremos que la dificultad del problema CDH (relativa a $\mathcal{G}$) implica dificultad del problema de logaritmo discreto (relativa a $\mathcal{G}$). Podemos hacer la demostración por contrarrecíproco. Es decir, queremos demostrar que si el logaritmo discreto es fácil, CDH también lo es. Asumimos que tenemos un adversario $\mathcal{A}_{DLOG}$ que resuelve el problema del logaritmo discreto con probabilidad no negligible (es decir, dados $(\mathcal{G}, q, g, h)$, puede devolver x tal que $g^x = h$). Luego, definimos un adversario $\mathcal{A}_{CDH}$ tal que hace lo siguiente.

1. Recibe $(\mathcal{G}, q, g, g^{x_1}, g^{x_2})$.
2. Obtiene $x_1 := \mathcal{A}_{DLOG}(\mathcal{G}, q, g, g^{x_1})$
3. Devuelve $(g^{x_2})^{x_1}$

Vemos que este adversario tiene probabilidad no negligible de ganar el experimento, ya que su probabilidad de éxito depende únicamente de la probabilidad (no negligible) de que $\mathcal{A}_{DLOG}$ tenga éxito.

Ahora, demostremos que la dificultad del problema DDH (relativo a $\mathcal{G}$) implica la dificultad del problema CDH (relativo a $\mathcal{G}$).

3. Suppose you know two El Gamal ciphertexts $\langle c_1, c_2 \rangle$ and $\langle c'_1, c'_2 \rangle$ that encrypt unknown plaintexts m and m' . You also know the public key h and cyclic group generator g .
 - (a) (5 points) What information can you infer about m and m' if you observe that $c_1 = c'_1$?
 - (b) (5 points) What information can you infer about m and m' if you observe that $c_1 = g \cdot c'_1$?
4. (10 points) In class we showed that El Gamal encryption is *malleable*, and specifically that given a ciphertext $\langle c_1, c_2 \rangle$ that is the encryption of some unknown message m , it is possible to produce a ciphertext $\langle c_1, c'_2 \rangle$ that is the encryption of $\alpha \cdot m$ (for known α). A receiver who receives both these ciphertexts might be suspicious since both ciphertexts share the first component. Show that it is possible to generate $\langle c'_1, c'_2 \rangle$ that is the encryption of $\alpha \cdot m$, with $c'_1 \neq c_1$ and $c'_2 \neq c_2$.
5. (10 points) Explain why the RSA exponents e and d must always be odd numbers.
6. (10 points) Consider the RSA-based encryption scheme in which a user encrypts a message $m \in \{0, 1\}^\ell$ with respect to the public key $\langle N, e \rangle$ by computing $\tilde{m} := H(m) \parallel m$ and outputting the ciphertext $[\tilde{m}^e \bmod N]$. (Here, let $H : \{0, 1\}^\ell \rightarrow \{0, 1\}^n$ and assume $\ell + n < \|N\|$.) The receiver recovers $\tilde{m}$ in the usual way and verifies that it has the correct form before outputting the ℓ least-significant bits as m . Prove or disprove that this scheme is CCA-secure if H is modeled as a random oracle.